

Chapter-1

Introduction to Database Systems



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Basic definitions

Data: Raw facts that can be recorded and do **not give full meaning** about objects.

- ▶ It simply exists and has no much significance beyond its existence.
- ▶ It can exist in any form.
- ▶ It does not have full meaning by itself.

Information: is a collection of related data that can give a **full meaning**.

- ▶ Processed data that have an explicit meaning.
- ▶ Need to understand the relation between data/datum.

Knowledge: an appropriate collection of information.

- ▶ Found through many experience and much organized information.
- ▶ Come through understanding patterns.



Cont....

Wisdom: collection of very deep knowledge.

- Come through understanding principles.





Database

- Is a **persistent**, logically coherent collection of inherently **meaningful data**, **relevant to** some aspects of the **real world**.
 - ▶ Is an **organized collection** of data
 - **Collection of related tables.**
 - ▶ **Collection of related data (logically coherent)** (Data items are related in a non-arbitrary way).
 - ▶ **Known facts that are recorded and that have a meaning.**
- Represents some aspects of the real world in the production environment.
- Mainly **built for a specific purpose**.

E.g: Amazon DB, Social media DB, different governmental DBs...



Database Management System (DBMS)

- A collection of **software** /programs that **facilitate** the **creation**, **manipulation** and **maintenance** of databases.
- **Allows multiple users** and programs **to access** and manipulate the database **concurrently**.
- **Provides protection** against **unauthorized** access and manipulation.

E.g:





Types of databases

- There are various types of databases used for storing different varieties of data:

Centralized Database

- It is the type of database that **stores data** at a **centralized database system**.
- It comforts the users to access the stored data **from different locations** through **several applications**.



Cont...

Advantages

- It has **decreased** the **risk of data management**
- Data **consistency** is **maintained** as it manages data in a central repository.
- It provides **better data quality**
- It is less costly because fewer vendors are required to handle the data sets.

Disadvantages

- The **size** of the centralized database is **large**
- It is **not easy to update** such an extensive database system.
- If any **server failure** occurs, **entire data** will be **lost**, which could be a huge loss.



Distributed Database

- **Data is distributed** among **different database** systems of an organization.
- These database systems are **connected via communication** links. Such links help the end-users to access the data easily.
 - **Examples:** Apache Cassandra, HBase, Ignite, etc.
- **Homogeneous DDB** and **Heterogeneous DDB**

Advantages

- One server failure will not affect the entire data set.



Relational Database

- This database is based on the **relational data model**, which **stores data** in the form of **rows**(tuple) and **columns**(attributes), and together forms a **table**(relation).
- A relational database **uses SQL** for storing, manipulating, as well as maintaining the data.
- Each table in the database **carries a key** that **makes the data unique** from others.
- **Examples** of Relational databases are MySQL, Microsoft SQL Server, Oracle, etc.



Properties of Relational Database

There are following four commonly known properties of a relational model known as **ACID properties**, where:

- **A means Atomicity:** This ensures the data operation will complete either with success or with failure. It follows the 'all or nothing' strategy. For Eg: a transaction will either be committed or will abort.
- **C means Consistency:** If we perform any operation over the data, its value before and after the operation should be preserved. For example, the account balance before and after the transaction should be correct, i.e., it should remain conserved.



Cont...

- **I means Isolation:** There can be **concurrent users** for **accessing** from the database. Thus, isolation between the **data should remain isolated**. For example, when **multiple transactions** occur at the **same time**, one **transaction effects should not be visible** to the **other transactions** in the database.
- **D means Durability:** It ensures that once it completes the operation and commits the data, **data** changes should remain **permanent**.



Reading assignment

- NoSQL Databases
- Cloud databases
- Object oriented databases
- Hierarchical databases
- Network databases
- Personal database
- Operational DB
- Enterprise DB
- End user database



Database vs. File-based Data handling/storage approaches

File-based:

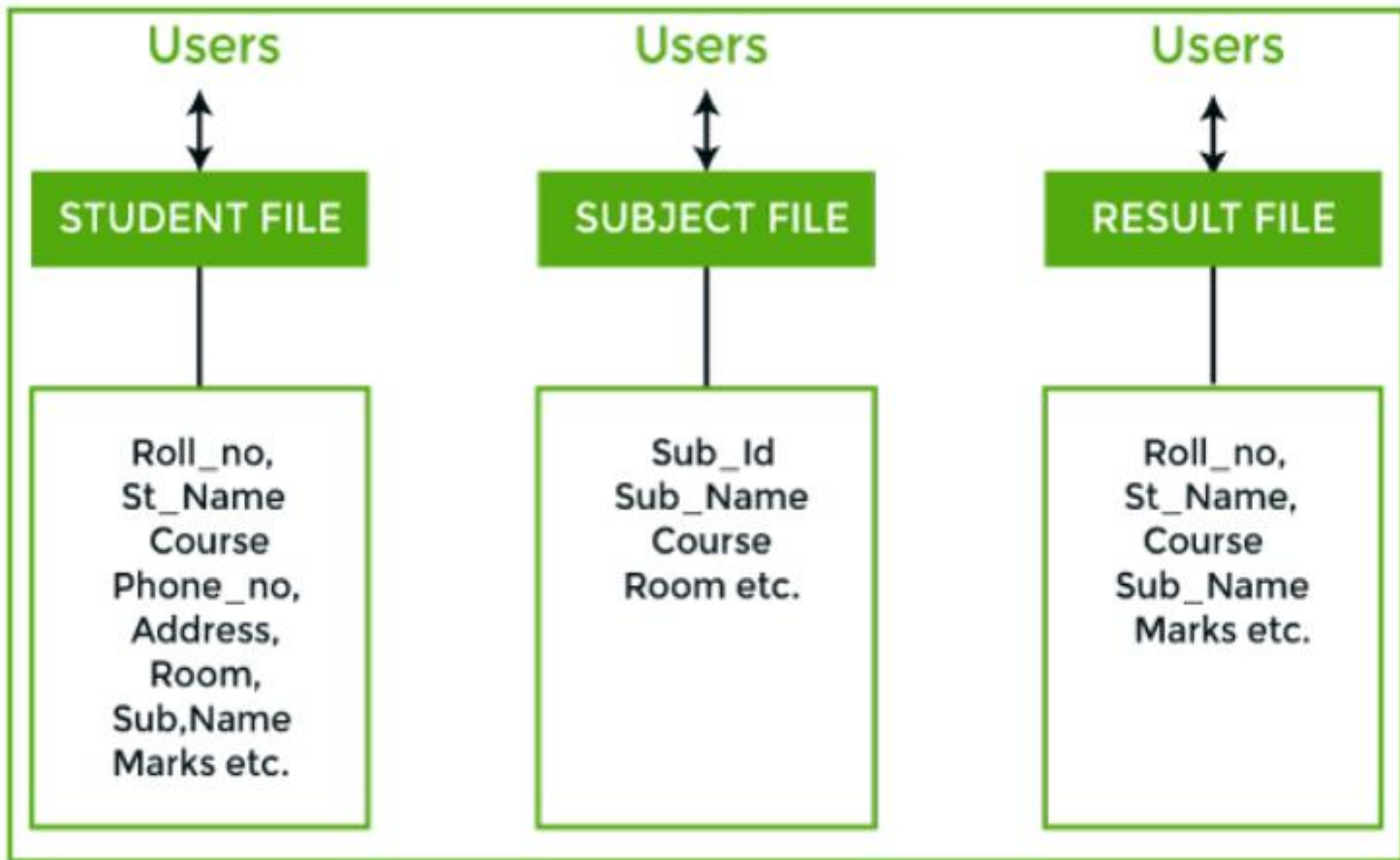
- ▶ The data is distributed in different files and can not be shared
- ▶ Data integrity checks are difficult
- ▶ Poor security
- ▶ Relatively less cost

DB-based:

- ▶ Data can be stored in one place and shared easily
- ▶ Data integrity and security are good
- ▶ Complex and costly relative to file based approach
- ▶ Recommended for the current growing systems

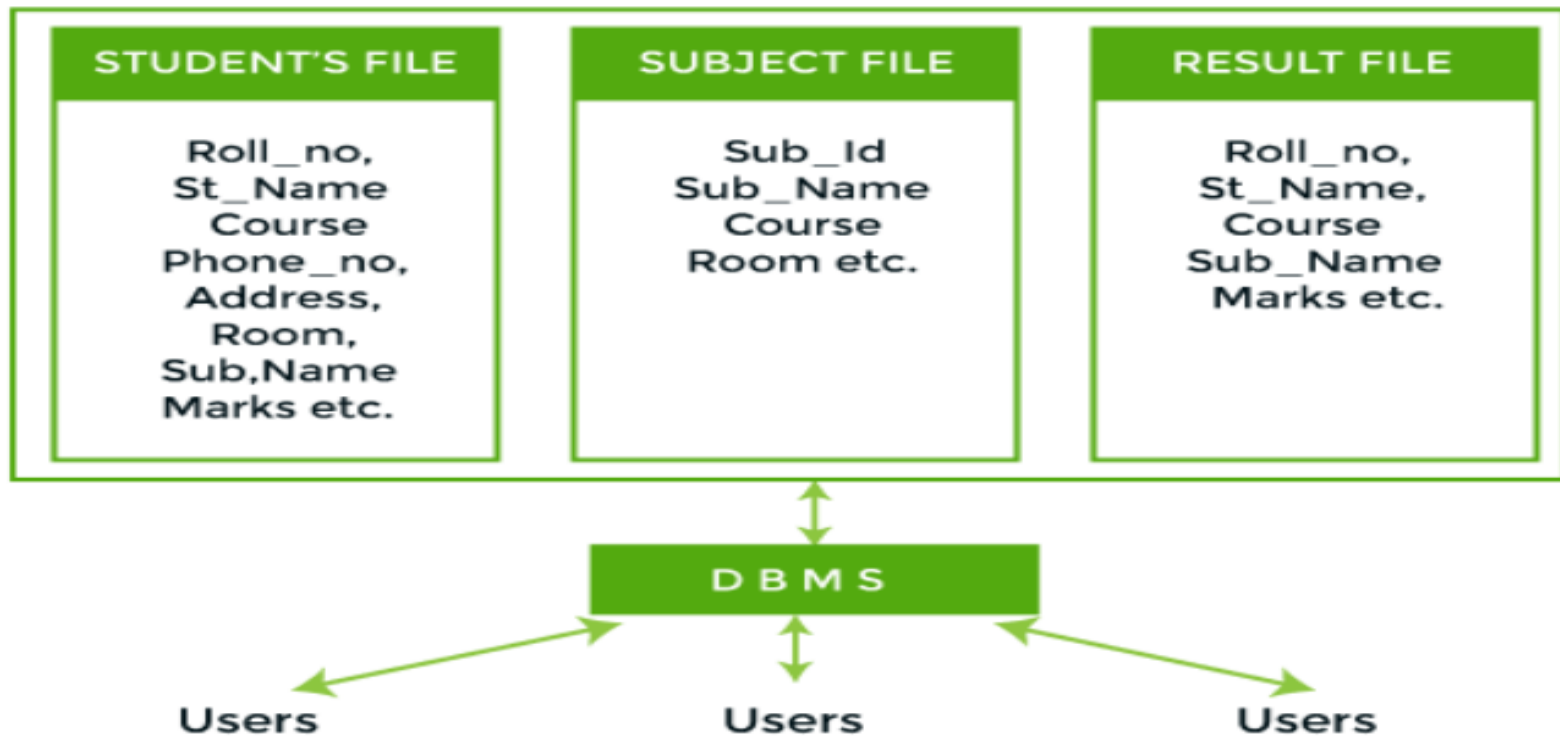


File-based approach





Database Approach





Limitations of File-based Approach

- Data redundancy
- Data inconsistency
- Unstandardized data
- Insecure data
- E.t.c



Why the Database Approach?

- Application needs constant change.
- Ad hoc questions need rapid answers.
- Need to reduce long development time and high cost in new application development.
- Lots of data shared throughout the organization.
- Need to improve data consistency and control access to data.
- Need to implement concurrency control.



Main Characteristics of the Database Approach

Self-describing:

- A DBMS catalog stores the description of the database.
- The description is called meta-data.

Insulation between programs and data (program-data independence)

- Allows changing data storage structures and operations without having to change programs.

Data abstraction: a data model is used to **hide storage details** and present the users with a conceptual view of the database.

Support of multiple views of the data: each **user** may see a **different view** of the database, which describes only the data of **interest** to that user.



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Sharing of data and multi-user transaction processing:

- Allowing a set of concurrent users to retrieve and to update the database.

Concurrency control:

- Within the DBMS guarantees that each transaction is correctly executed or completely aborted.

OLTP (Online Transaction Processing):

- Is a major part of database applications.



Database Users/Actors

Database users may be divided into:

Actors on the Scene

- Those who actually use and control the content

Actors/Workers behind the Scene

- Those who enable the database to be developed and the DBMS software to be designed and implemented

Note: Technical individuals such as designers, analysts, programmers, tool developers and sometimes database administrators can sometimes work both at the back and front end.



Actors on the scene

Database administrators:

- Administering primary and secondary database resources
- Authorizing access to the database
- Acquiring software/hardware resources,
- Controlling its use and monitoring run-time performance.
- Installing and upgrading the database server and/or application tools.
- Creating a backup and recovery strategy for the database

Application programmers:

- know how to interact with the system but may not know how DBMS is designed.
- Test, debug, document and maintain transactions

System analysts: determine and analyze the requirements of end users



Cont...

Database designers:

- Identifying the data to be stored in the database and for choosing appropriate structures to represent and store the data.
- These tasks are mostly undertaken before the database is actually implemented and populated with data.
- It is the responsibility of database designers to communicate with all prospective database users. (understand their requirements and design the database)



Cont...

End-users: use the data for queries, reports and some even update database content.

- **Casual end-users:** occasionally access the database, but they may need different information each time.
- **Naïve/ parametric:** do not know about database too much, invoke application programs that are prepared already.
- **Stand-alone end-users:** maintain personal databases by using easy-to-use ready-made program packages.
- **Sophisticated users:** know advanced use of the system and packages on the top of the system. Eg: engineers, scientists, business analysts who are familiar with DBMS



Actors behind the scene

Operators and maintenance personnel

- Responsible for actual running and maintenance of hardware and software

Tool developers

- Persons who design and implement tools (software package)

DBMS designers and programmers

- Designs and implement DBMS modules & interfaces as a package.



Advantages of Using the Database Approach

- More information from given database
- Ad-hoc queries can be performed
- Redundancy can be reduced
- Inconsistency can be avoided
- Security restriction can be applied
- Data independence/isolation
- Reduced development time, flexibility.
- Sharing of data among multiple users.
- Provide persistent storage for program objects (in Object-oriented DBMS's).
- Provide storage structures for efficient query processing.



Cont...

- Provide backup and recovery services.
- Provide multiple interfaces to different classes of users.
- Represent complex relationships among data.
- Enforce integrity constraints on the database.



Disadvantage

- **Expensive:** sometimes hardware, software, personnel, processing overhead, operating cost , etc. may be high.
- Increased **vulnerability** to attack
- **Failure**(sometimes).
- Recovery may become more **complex**.



Database Architecture: ANSI-SPARC (American National Standards Institute, Standards Planning And Requirements Committee)

Internal schema:

- Describes the **physical storage structure** of the database.
- This uses a physical data model and describes the complete details of data storage and access paths for the database.

Conceptual schema:

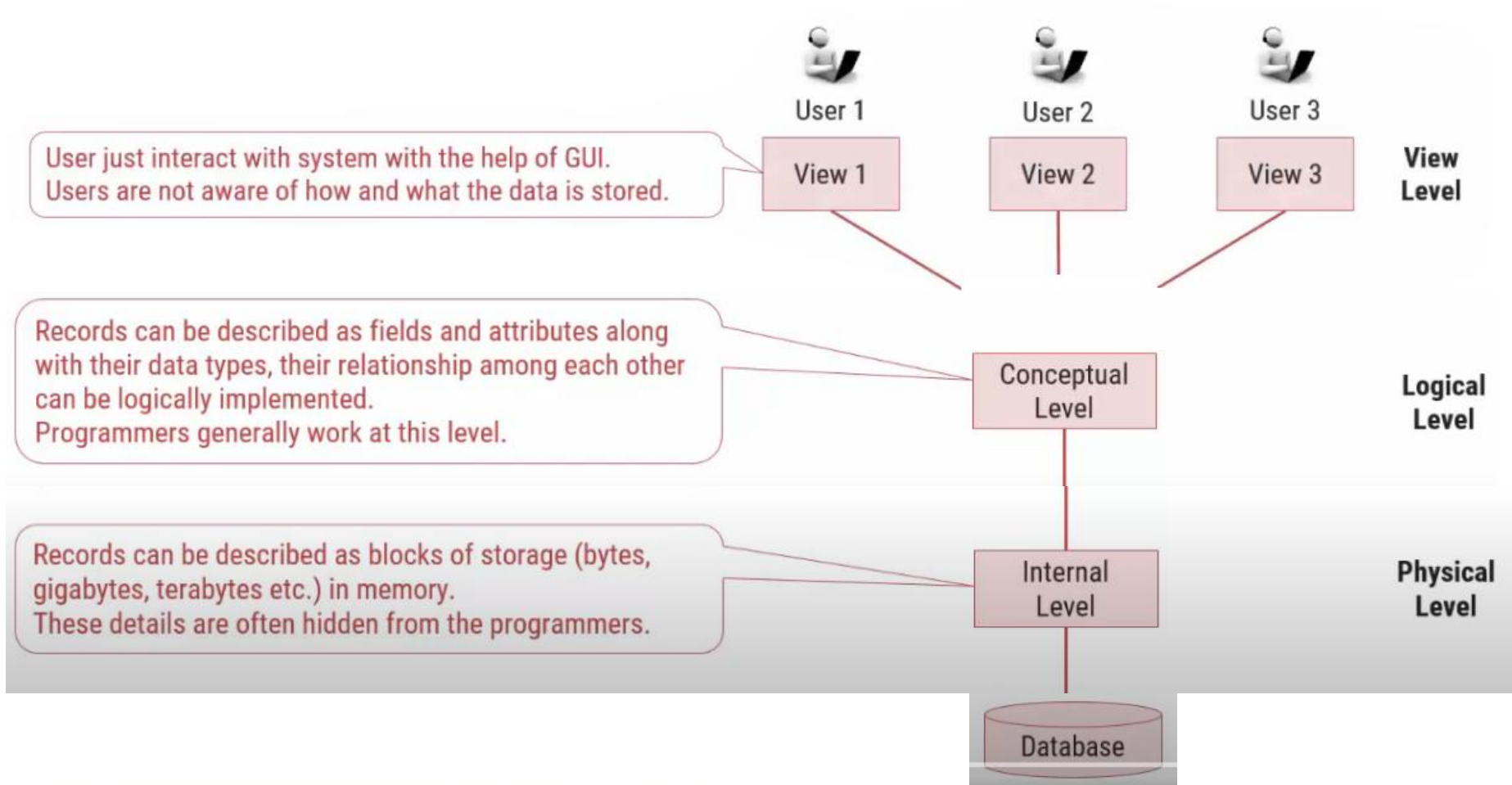
- Describes the structure of the whole database for a community of users
- It hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations and constraints at a very high level.
- The implementation of conceptual schema is often based on a conceptual schema design in a high level data model.

External schemas or user views:

- Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from the other user group.

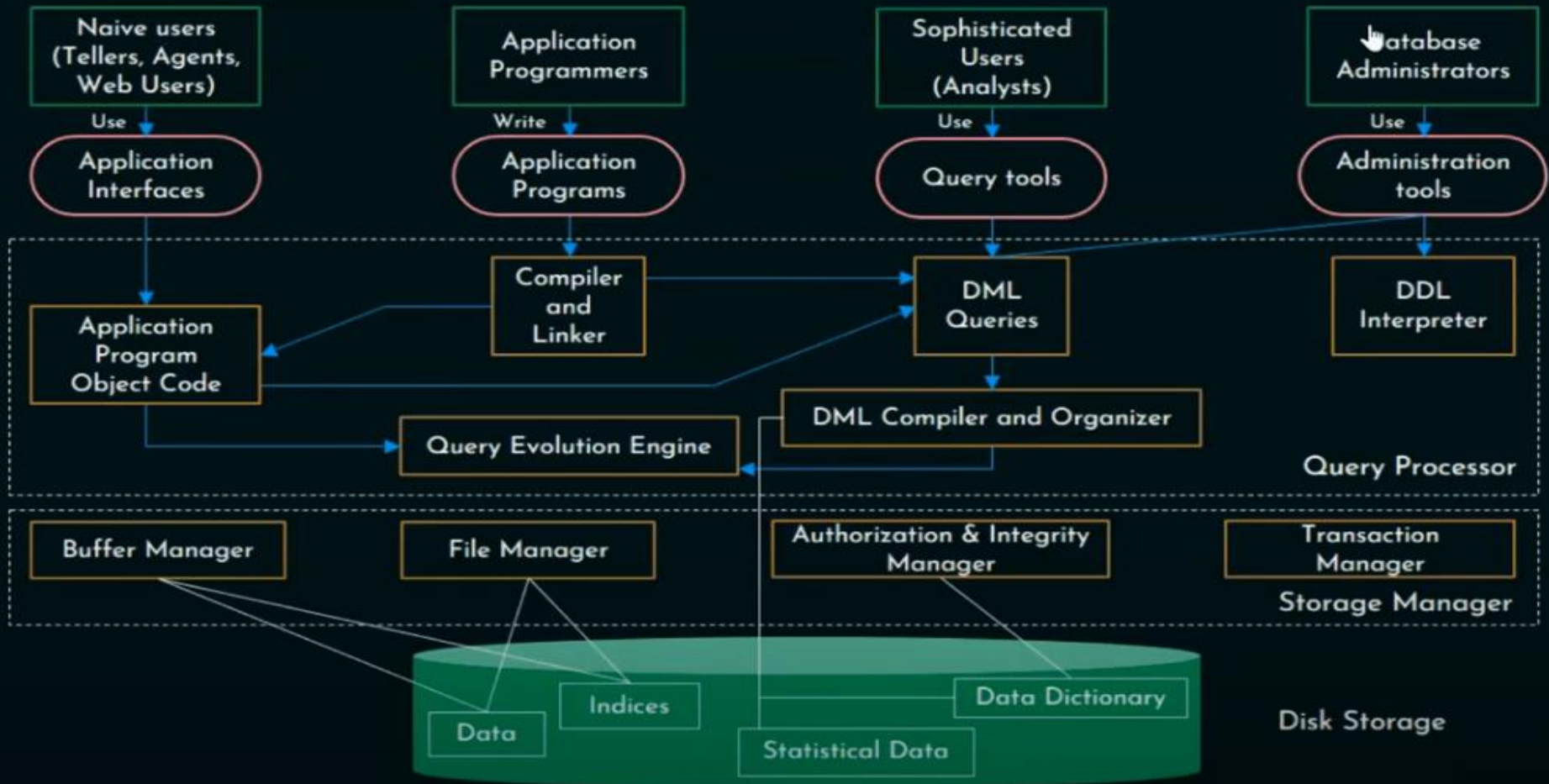


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Detail diagram





Reading assignment

- Functions/ Applications of DBMS



Database Languages

- A DBMS has appropriate languages and interfaces to express database queries and updates.
- Database languages can be used to read, store and update the data in the database.

Types of database languages

1. Data Definition Language (DDL)
2. Data Manipulation Language
3. Data Control Language (DCL)
4. Transaction Control Language (Transaction Control Language)



Data Definition Language (DDL)

- A language which specify the database schema.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- Used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.
- E.g: CREATE, DROP, ALTER, RENAME, TRUNCATE



Cont..

- **Create:** It is used to **create objects** in the database.
- **Alter:** enables you to **change the overall** characteristics of a database.
- **Drop:** It is used to **delete objects** from the database.
- **Truncate:** It is used to **remove all records** from a table.
- **Rename:** It is used to rename an object.
- **Comment:** It is used to comment on the data dictionary.



Data Manipulation Language (DML)

- It is used for accessing and manipulating data in a database.
- It handles user requests.
- **Select:** It is used to retrieve data from a database.
- **Insert:** It is used to insert data into a table.
- **Update:** It is used to update existing data within a table.
- **Delete:** It is used to delete all records from a table.
- **Merge:** It performs UPSERT operation, i.e., insert or update operations.
- **Call:** It is used to call a structured query language or a Java subprogram.
- **Explain Plan:** It has the parameter of explaining data.
- **Lock Table:** It controls concurrency.



Data Control Language

- It is used to retrieve the stored or saved data.
- The DCL execution is transactional. It also has rollback parameters.
 - **Grant:** It is used to give user access privileges to a database.
 - **Revoke:** It is used to take back permissions from the user.



Transaction Control Language (TCL)

- TCL is used to run the changes made by the DML statement. TCL can be grouped into a logical transaction.
- Here are some tasks that come under TCL:
 - **Commit:** It is used to save the transaction on the database.
 - **Rollback:** It is used to restore the database to original since the last Commit.



Thank you !!!